

Battery safety instructions

Avant Power OptiTemp® Li-ion batteries

OptiTemp® 13 kWh A458935

OptiTemp® 13 kWh Retrofit A466799

OptiTemp® 27 kWh A458226



WARNING



Risk of serious injuries and fire – Always handle the battery carefully. Read this document whenever handling the in any way a battery that is separated from a loader, including the following situations:

- storage
- transport of a battery
- replacing a battery

Damaged battery, wrong handling, or misuse can cause battery to rupture, releasing chemicals from the battery, or fire.

These instructions are intended only for handling of a battery that is separated from an Avant loader.

For instructions about using, charging, and handling a battery that is installed to a loader always refer to the Operator's Manual of the loader.

1. SAFE HANDLING OF OPTITEMP LI-ION BATTERY

1.1. BASIC DESCRIPTION OF SAFETY AND INSTALLED REDUNDANCY

- Cell level
 - Safety devices in the top of the battery pack housing can prevent a cell from thermal runaway or explosion in the abusive environment, including overcharging and external short circuit.
- Module level
 - Special assembly pattern with a safety distance to prevent a chain reaction between cells resulting from a hazardous situation.
 - Each module has a control unit for monitoring and management of the complete module including the cell voltage and temperature.
- Battery system level
 - Redundant Battery Management System (BMS) monitoring and managing all the modules and interface to the Loader and chargers, protecting the battery system from abuse. With the constant monitoring of the battery system any situation outside normal operating parameters (i.e. higher or lower temperature, or higher or lower voltage) will generate error codes according to different escalations and take action by for example switching off loader completely to prevent hazardous situations.
 - Mechanical protection to prevent penetration, shock and vibration.

1.2. EXAMPLE CASES OF UNUSUAL SITUATIONS

- Case 1
 - The mechanical protection of the battery system is penetrated (e.g. forklift forks through the battery)
 - See page 5 - Firefighting measures
- Case 2
 - Loader on fire
 - See page 5 - Firefighting measures

1.3. LI-ION BATTERY SYSTEM CONFIGURATION

44 V battery	Module Ah _{nom}	Module qty	Cell qty	Capacity
Double-module OptiTemp battery	260 Ah	2	720	13 kWh
4-module OptiTemp battery	520 Ah	4	1440	27 kWh

2. HAZARD IDENTIFICATION

2.1. CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

No risks occur during normal use of lithium-ion batteries when handled in accordance with the operator's manual supplied with the loader.

However, if exposed to fire, subjected to mechanical shock, disintegration or incorrectly connected / charged, the hazardous substances of the battery can be released, resulting in a risk of fire or harmful gas may be generated.

2.2. OTHER HAZARDS

None identified.

2.3. LABEL ELEMENTS

In accordance with the EU batteries directive and current national legislation, lithium-ion batteries are labeled with a crossed bin. Other labels: see Operator's manual.



2.4. BATTERY SUBSTANCES

As a solid, manufactured article, exposure to hazardous ingredients is not expected during normal use.

3. PHYSICAL AND CHEMICAL PROPERTIES

Note! Read carefully material safety datasheets attached with loader technical file:

- A466139 EN OptiTemp Fluid - SDS

The following applies to electrolyte leaks:

Proper actions

Use general ventilation or spot extraction to keep exposure below hygiene limits. Avoid direct contact with the electrolyte. Emergency shower and eye shower should be available in the workplace.

Respiratory protection

In case of insufficient ventilation, use respiratory protection adapted for the specific conditions of use.

Skin / body protection

Use chemical-resistant gloves. Use full protective clothing and chemical-resistant apron.

Eye protection

Wear safety glasses.

Information on basic physical and chemical properties of the battery housing

The battery consists of an outer sealed housing which is made of polypropylene and polyethylene plastic.

Form	Solid
Colour	Green/black
Odour	Odourless
pH value	Not applicable
Flash point	Not applicable
Lower explosion limit	Not applicable
Vapour pressure	Not applicable
Density	Not applicable
Water solubility	Insoluble
Ignition temperature	Not applicable
IP class	IP54

4. TOXICOLOGICAL INFORMATION

If appropriately handled and in accordance with the general safety rules, no health hazards have become known. However, the product contains substances that are considered to be hazardous to health.

Information on toxicological effects

Acute toxicity	LD ₅₀ oral > 2000 mg/kg or more
Irritating nature	Irritative to skin and eyes
Mutagen	Not applicable
Chronic effects of overexposure	Not applicable

5. ECOLOGICAL INFORMATION

Ecological injuries are not known or expected under normal use. There is no mercury (Hg), cadmium (Cd) or lead (Pb) in the battery.

6. DISPOSAL CONSIDERATIONS

Waste treatment methods

Batteries should be handled as hazardous waste and shall be recycled in special facilities approved for the purpose. Spent lithium-ion batteries are subject to regulation of the EU Battery Directive (2006/66/EC) and its adoption into national legislation on the composition and end-of-life management of batteries. Cleaned packaging material is recycled in accordance with national regulations.

7. HANDLING AND STORAGE

Precautions for safe handling

- Avoid short circuit.
- Avoid reverse polarity (incorrect connection) during installation.
- Keep the battery dry.
- Do not expose the battery to strong oxidizing agents.
- Do not damage or remove the battery cover.
- Keep away from open flames, hot surfaces and sources of ignition.
- Do not solder directly on the battery.
- Do not damage or deform the battery.
- Protect the poles from mechanical impact.
- Use only chargers intended by the manufacturer.

Recommendations for safe storage, including any incompatibilities

- Do not store the batteries in dusty areas and avoid water, acid and oxidising agents.
- Store indoors, between 0 °C and +25 °C in a dry environment, air humidity below 70 %.
- Keep battery SOC at 50-75 %. Optimum temperature is just below 20 °C.
- Avoid direct sunlight, high temperature and high humidity.
- Keep in closed container to prevent short circuit.
- Check charge level every three months.

8. EXPOSURE DURING OPERATION

Control parameters

Not specified. During normal charging and discharging no release of any hazardous substances occur.

Exposure controls

During normal charging and discharging no release of any hazardous substances occur.

9. STABILITY AND REACTIVITY

Reactivity

Not expected to be reactive. In case of leakage, the electrolyte reacts with water.

Chemical stability

Stable under normal conditions of use and storage.

Possibility of hazardous reactions

No dangerous reactions are known.

Conditions to avoid.

- Short circuit.
- Mechanical impact or crush that can cause deformation.
- Temperature above 80 °C.
- Direct sunlight.
- High humidity.

Incompatible materials

- Electrically conductive material.
- Water, metal pieces.
- Strong oxidizing agents.
- Strong acids.

Hazardous decomposition products

No decomposition occurs during storage and use as instructed.

In case of fire and open cells, there is a possibility of irritation or release of harmful gases.

9.1. COLD STORE OPERATIONS

If the loader is used for Cold Store use, it is possible to work in cold store areas for a limited time.

For continuous cold store operations, the battery is equipped with heating as a default. The battery will then keep itself warm by internal heating elements. The heated batteries can be charged in cold store, but the charger must be kept warm by an optional housing or placed outside. If used onboard charger, check required operation temperature from the loader Operator's manual.

10. FIRST-AID MEASURES

The following first-aid measures are required only in case of exposure to interior battery components after damage of the external battery housing. Undamaged, closed cells do not represent a danger to the health.

EXPOSURE

After inhalation

After contact with skin

After contact with eyes

After ingestion

ACTION

Ensure fresh air. Seek medical attention.

Wash off immediately with plenty of water. Remove all contaminated clothing. If skin irritation persists, consult a physician.

Rinse immediately with plenty of water for at least 15 min, also under the eyelids, for at least 15 minutes. Seek medical treatment by eye specialist.

Wash out mouth thoroughly, do not induce vomiting unless instructed by medical personnel. Call a physician immediately.

11. FIREFIGHTING MEASURES

If you discover fire or smoke coming from the loader while driving, switch off the loader and step away from the danger area.

If you discover fire or smoke while charging the loader, switch off the charging unit or disconnect it and step away from the danger area.

11.1. SMOKE FROM THE BATTERY

Important countermeasures:

- Bring the loader outside if possible
- Cool down the battery thermal leakage, water can be used
- Avoid inhaling the smoke

11.2. BATTERY OR CHARGER ON FIRE

Follow these steps:

1. Vacate the room
2. Call the fire brigade
3. Cut the main power to the charger and disconnect the battery charger plug
4. Start the firefighting according to the recommendations below
 - Use plenty of water / carbon dioxide gas / nitrogen gas / chemical powder / fire foam extinguisher aimed directly at the fire
 - When the battery fire is extinguished, keep cooling down the battery with water until the fire brigade arrives
 - If the fire is started in an enclosed space, it could be extinguished through an automatic fire sprinkler or manual controlled sprinkler system.

12. ACCIDENTAL RELEASE MEASURE

This information is relevant only if the battery is broken and the contents are released.

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Avoid contact with skin, eyes and clothing. Avoid breathing fume and gas.

Environmental precautions

Do not discharge into the drains/surface waters/groundwater.

Methods and material for containment and cleaning up

Take up mechanically and send for disposal. The solids are put in a container.

Handling of broken batteries

In any case of malfunctioning or damage to the battery, park the machine in a safe place and contact Avant service representative.

See section 3 page 2 for personal protection and section 6 page 3 for Disposal considerations.

13. SAFETY PRECAUTIONS AND RECOMMENDATIONS FOR CHARGING AREAS

The site owner is responsible for the installation and precautions around the charging areas for Lithium-Ion batteries. Always check with your insurance company for specific requirements or if any local regulations apply to loaders with Li-ion batteries. Avant always recommend making a local risk assessment of the installation to meet the local regulations regarding electrical high-power equipment for industrial use. As the Avant Lithium-Ion batteries don't emit any gas during the charging process, the charging areas can be placed anywhere in the building without mechanical ventilation, following our recommendations below.

Note: The high-power Li-Ion chargers might emit a lot of heat during their normal operation.

General recommendations for selecting the area and precautions for charging Li-Ion batteries:

Installation

- Select an area without heavy traffic or install barrier protections for the vehicles put on charge
- Don't place the charger in a small, closed room without any air circulation
- Dusty areas or premises with high humidity should be avoided as charging areas
- The floor, walls and ceiling must be made of fire safe material
- The chargers should be steady mounted on the wall at >1500 mm height from the floor, >300 mm from the ceiling and with a space of >200 mm to the next charger
- Remove all flammable material around the chargers
- The charger must be possible to unplug from the mains or disconnected by an all-pole power switch

Handle & Care

- Check that the charger plug is fully inserted to the battery and locked before charging

- Handle the charging plug and cable with care, never drop them on the floor or pull the cable too hard
- Make regularly inspections on cables and sockets, replace at any signs of wear
- If the charger gets any mechanical damages, disconnect mains plug and repair before use
- Make sure the Avant Safety label about safe Li-ion charging is attached, see Operator's Manual A465990 section 4.1, for more information

Fire precautions

- Fire extinguisher of type AVD/10kg should be mounted accessible near the charging area
- Equip the charging area with a smoke & fire detection alarm
- Preferable also install a water tap & hose close to the charging area
- If possible, install an automatic fire sprinkler system if the charging takes place during unmanned hours

Read carefully the instructions in the Operator's Manual for the loader and charger.

13.1. CHARGING INFORMATION

During the charging process the Li-ion battery controls charger and the power supply.

The battery BMS secures that the charger is switched off at any malfunctions or high or low temperatures.

The recommended temperature range for charging Li-ion loaders is under 35 °C. Under 10 °C the charging speed is reduced.

Avant charging connector

The Avant Li-ion charging connector has following features:

- The electrical safety switch system secures that the power is cut before disconnected
 - no risk for sparks
- LED-indications on plug indicates the charging process
- Ergonomic shape ensures smooth handling for frequently use
- Easy disconnection via release button
- Dust-protected and durable design

14. TEST AND CERTIFICATION

Avant Li-ion batteries are assembled along our production line in our factory according Avant Power's quality system.

Several safety diagnostic systems are installed to ensure highest safety levels can be achieved e.g temperature and voltage measurement on the modules and the complete battery. If something is wrong down to cell level, the battery is shut down. Each battery cell has an overtemperature valve that releases the pressure if the cell becomes overheated. All cells are arranged with a small space in-between, to prevent a chain reaction if one cell gets overheated.

Multiple tests are made to ensure battery safety and risk assessment:

- Altitude simulation
- Thermal shock cycling
- Temperature
- Vibration
- Mechanical shock
- External short circuit
- Overcharge

- Overdischarge
- Immersion
- Collision
- Drop
- Propagation

OptiTemp li-ion battery system meets or exceeds regulation requirements.

With the “EC Declaration of Conformity” that is provided with each loader, Avant takes full responsibility of the products and declares that they meet or exceed all requirements of the applicable EU Directives.

Other documentation from the Technical File are considered as confidential and must not be disclosed externally.

15. CERTIFICATE OF CONFORMITY

The Battery System is a part of the loader’s CE marking according to the applicable regulations.

16. NORMS & REGULATIONS

- MD 2006/42/EC machine directive (EU)
- EMC 2014/30/EC electromagnetic compatibility (EU)
- BD 2006/66/EC Battery Directive (EU)
- REACH No 1907/2006 (EU)

Certificate	Status
IEC 62619	Fulfilled
UN 38.3 7th Ed.	Fulfilled
IEC 62281	Fulfilled
IEC 62485-6	Fulfilled
EN ISO 12100	Fulfilled
EN ISO 13849	Fulfilled

17. TRANSPORT INFORMATION

The battery is always transported as part of the loader. If the battery is transported separately following rules must be followed:

- Safety documentation must be included.
- After transportation, check that the delivery is complete and not damaged.
- Replacement or separate battery components should only be transported in coordination with dedicated service following the mandatory rules about packaging, labelling and documents which apply for each transport method.

Transport body	UN Number	UN shipping name	Hazard class	Packing group
ADR/RID	UN 3480	Lithium-ion batteries	9	II

IMDG	UN 3480	Lithium-ion batteries	9	II
IATA/ICAO	UN 3480	Lithium-ion batteries	9	II
IMDG	UN 3171	Battery-powered vehicle	9	Battery attached in loader

Recommendations for safe transportation of Li-ion Loaders

If possible, reduce the Li-ion battery charging level to <50 % before road transports. The risk in case of an accident will be reduced in relation to the stored energy level. The ambient temperature range during transportation should be within -20 °C to +40 °C. If the battery temperature is below 0 °C after transportation, the loader must be stored indoor temperature to acclimatize the battery before normal use. If the vehicle is used with a battery temperature between 0 to 10 °C, the performance of the loader and functions are limited to protect the cold battery. Below battery temperature 0 °C the machine can't be started.

Special precautions for user

Not applicable.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

Not applicable.

18. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture.

In accordance with the EU Battery Directive and the respective national legislation, lithium-ion batteries have to be marked by a crossed-out dust bin. Lithium-ion batteries are regulated by the EU Battery Directive (2006/66 / EC) and its implementations in national legislation on the composition and management of end-of-life batteries.

19. OTHER INFORMATION

The information describes the safety requirements for the products and is based on the present level of our knowledge. Data of all sections do not necessarily refer to the use and the regular handling of the product (always refer to the operator's manual for adequate product use). This data does not constitute a guarantee for the characteristics of the products as defined by the legal warranty regulations. The data for the hazardous ingredients were taken respectively from the last version of the sub-contractor's safety data sheet.